

Week 2: Geometric Reasoning

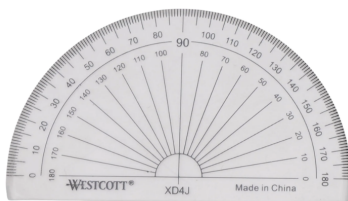
Jacob Khohayting

Last updated: September 20, 2026

Contents

360°	2
Vertical Angles	2
Transversals	3
Angles in a Triangle	4
The Boss: Right Triangles	4
Extending to Polygons	5
Practice Problems	6

※ 360°



Above is an image of a protractor. Do you know how a protractor works? How much degrees are in a line? How much degrees are in a full rotation?

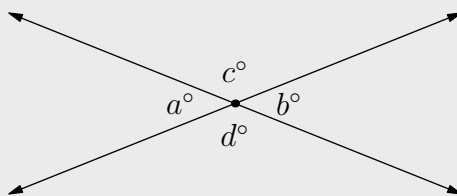
A protractor is a tool used to measure and draw angles. It consists of a flat, straight base and a semi-circular curved edge marked with degrees from 0 to 180. To measure an angle, the central point at the bottom of the protractor is placed directly on the vertex (the corner) where the two lines meet. One line of the angle is aligned with the baseline at 0° , and the measurement is read where the second line crosses the curved scale. The protractor features two sets of numbers running in opposite directions so that angles opening from either the left or the right can be easily measured.

The bottom edge of the protractor forms a perfectly straight line. If you track the measurements from one end of this straight edge to the other, the numbers span from 0° to 180° . Therefore, there are exactly 180 degrees in a straight line.

A full rotation represents a complete circle. Because a standard protractor is a semicircle containing 180 degrees, a full circle is equal to two protractors joined together. Multiplying 180 degrees by two results in 360 degrees in a full rotation.

※ Vertical Angles

Suppose two lines ℓ_1 and ℓ_2 intersect to form four angles with measures a° , b° , c° , and d° . If $a = 48$, find b , c , and d ?



Recall, a straight line sums to 180° ! So,

$$a + c = 180$$

$$c + b = 180$$

$$b + d = 180$$

$$d + a = 180$$

Since $a = 48$, from the first equation, $c = 180 - 48 = 132$. Then, from the following equation, $c + b = 180$ so $b = 48$. Then, finally $d = 132$ from the third equation. We can check this satisfies the fourth equation.

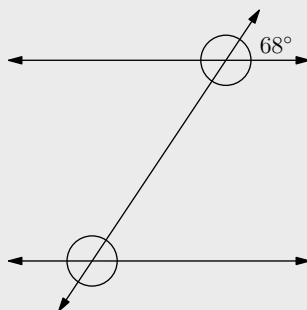
Notice that $a = b$ and $c = d$! We call this the **Vertical Angles Theorem**: When two lines intersect, the angles that are directly opposite each other (across the intersection point) are called vertical angles. This theorem states that vertical angles are always equal to each other.

Just by looking at the diagram, this does make sense.

✂ Transversals

A transversal is a line intersecting two parallel lines.

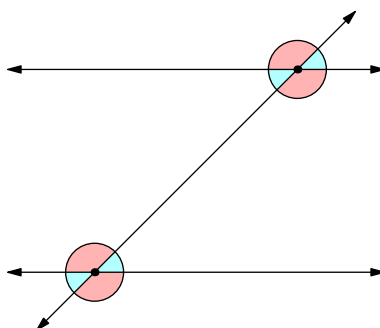
Two parallel lines intersect a third line. One angle measures 68° , as shown below. Can you fill in the remaining seven angle measurements?



Notice, the four angles in the higher horizontal line can all be found in a similar method to the Vertical Angles Problem! I'll leave that to you.

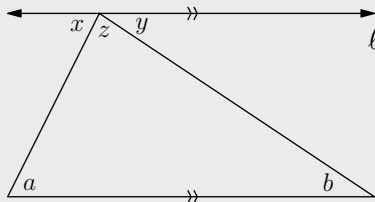
What about the bottom horizontal line. Well, because the lines are parallel, the angles in the exact same positions match perfectly. In a way, we could simply shift the top line down to the bottom line. From, there we get the angles are the same!

Some of these angle relationships have names that are too annoying so we won't learn them today. All you need to know is the following: in the following diagram, all red angles are the same and all blue angles are the same. In addition, all red angles + blue angles sum to 180° .



※ Angles in a Triangle

Find the sum of the angles in a triangle. Use the below diagram to help you! The line is parallel to the bottom side of the triangle. What angles are equal? What equations can you make?



Notice, $a = x$ and $b = y$ because of the properties of transversals. In addition, $x + y + z = 180^\circ$ because they all lie on a straight line. So,

$$x + y + z = a + b + z = 180^\circ.$$

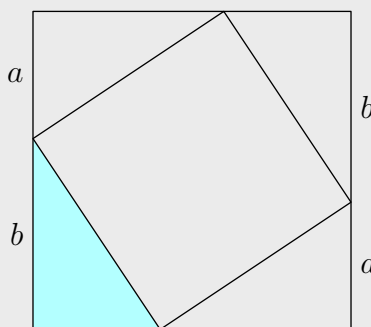
Incredible, we have found the sum of the angles is indeed 180° !

In summary, the total measure of all three interior (inside) angles of any triangle always adds up to exactly 180° . This rule is universal, meaning it applies to tiny triangles, giant triangles, and triangles of any shape!

※ The Boss: Right Triangles

Recall, the area of a square is s^2 .

Consider a large square with side length $(a + b)$. Find the area of the smaller square in terms of a and b . Can you formulate a final statement?



The large outer square has a total side length of $(a + b)$. This means its total area is:

$$\text{Area}_{\text{Total}} = (a + b)^2 = a^2 + 2ab + b^2$$

Inside this large square, we see four identical right-angled triangles (shaded in the original diagram). Each triangle has a base of a and a height of b (or vice versa). The area for one of these triangles is $\frac{1}{2}ab$.

Since there are four of them, their combined area is:

$$\text{Area}_{4 \text{ Triangles}} = 4 \times \left(\frac{1}{2} ab \right) = 2ab$$

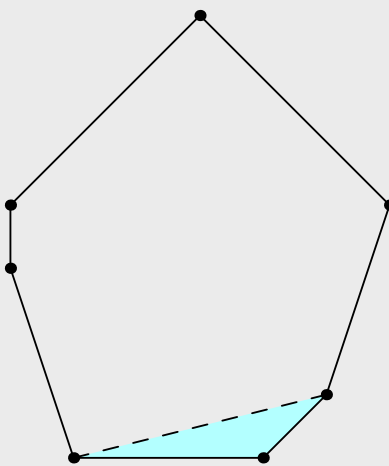
The area of the smaller, tilted square is simply the "leftover" space when you subtract those four triangles from the large outer square:

$$\text{Area}_{\text{Inner}} = \text{Area}_{\text{Total}} - \text{Area}_{4 \text{ Triangles}} = a^2 + b^2.$$

As a result, had the inner square have side length c , then $a^2 + b^2 = c^2$. Therefore, for each of the smaller right triangles, this property holds. Hence, for a right triangle with legs a and b along with hypotenuse c , then $a^2 + b^2 = c^2$, as desired.

✂ Extending to Polygons

Consider any arbitrary 7-gon, such as the one below:



What is the sum of the angles in the heptagon? Use a technique of forming triangles to help you! Try to use what we found in the previous section.

Now that we know every single triangle in the universe has inside angles that add up to exactly 180° . We can use this by noting any n -gon can be cut into $n - 2$ triangles. Each triangle has a sum of angles of 180° . So, the sum of the angles in a polygon is $180^\circ \cdot (n - 2)$.

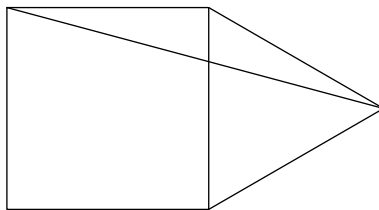
We will look at more of this next week!

✧ Practice Problems

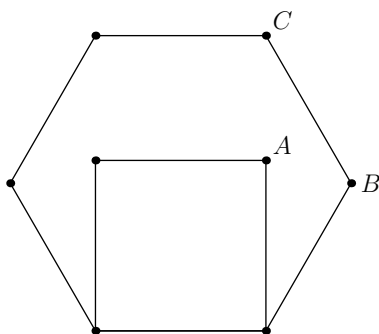
1. Find the ratio of the side lengths of the following types of triangles:

- Triangle with angles 45° , 45° , and 90° . This is called an *isosceles right triangle*.
- Triangle with angles 60° , 60° , and 60° . This is called an *equilateral triangle*.

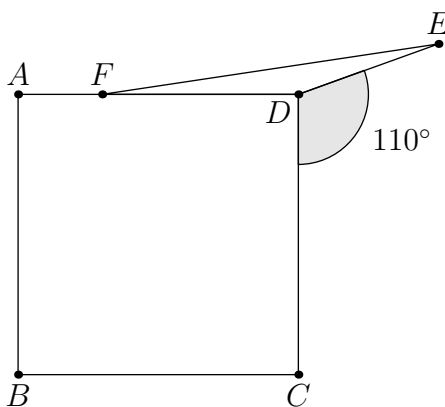
2. In the adjoining figure, $ABCD$ is a square, ABE is an equilateral triangle and point E is outside square $ABCD$. What is the measure of $\angle AED$?



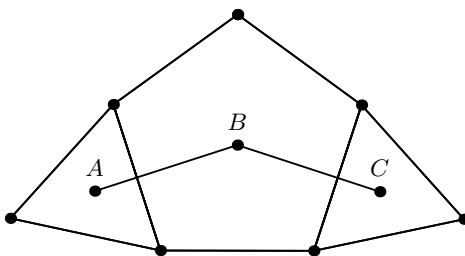
3. A square and a regular hexagon share a side. What is the measure of $\angle ABC$?



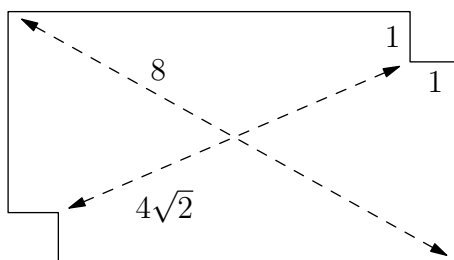
4. As shown in the figure below, point E lies on the opposite half-plane determined by line CD from point A so that $\angle CDE = 110^\circ$. Point F lies on $\overline{AD}$ so that $DE = DF$, and $ABCD$ is a square. What is the degree measure of $\angle AFE$?



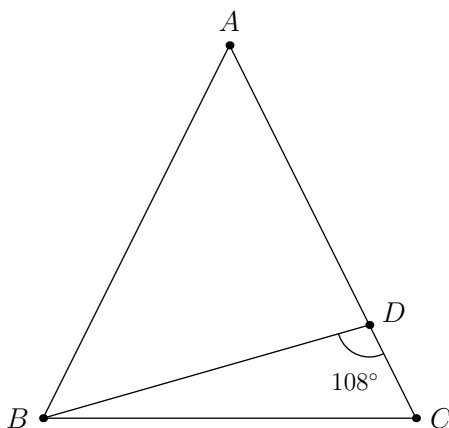
5. The figure below is formed by attaching two equilateral triangles along the edges of a regular pentagon. When their centers A , B , and C are connected, what is the measure of the angle $\angle ABC$?



6. Let $\triangle ABC$ be an equilateral triangle and P a point on $\overline{BC}$. If $PB = 50$ and $PC = 30$, compute PA .
7. On a rectangular page $ABCD$, Celine marks point P lies on segment $\overline{AC}$ such that $AP = 10$. If the distances from P to sides $\overline{BC}$ and $\overline{CD}$ are 3 and 4, respectively, what is the area of $ABCD$?
8. Daniel finds a rectangular index card and measures its diagonal to be 8 centimeters. Daniel then cuts out equal squares of side 1 cm at two opposite corners of the index card and measures the distance between the two closest vertices of these squares to be $4\sqrt{2}$ centimeters, as shown below. What is the area of the original index card?



9. In isosceles triangle ABC with $AB = AC$, let D be the unique point on side $\overline{AC}$ for which $AD = BD$. If $\angle BDC = 108^\circ$, what is the measure of $\angle BCD$?



10. Angle ABC of $\triangle ABC$ is a right angle. The sides of $\triangle ABC$ are the diameters of semicircles as shown. The area of the semicircle on $\overline{AB}$ equals 8π , and the arc of the semicircle on $\overline{AC}$ has length 8.5π . What is the radius of the semicircle on $\overline{BC}$?

